DISCRETE MATHEMATICS
GROUP CODES

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INTRODUCTION TO CODING THEORY

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•The process of communication involves transmitting some information carrying signal(message) that is conveyed by a sender to a receiver.

•Even though the sender may like to have his message received by the receiver without any distortion ,it is not possible due to a variety of disturbances(noise) to which the communication channel is subjected.

•Coding theory deals with minimizing the distortions of the conveyed message due to noise and to retrieve the original message to the optimal extent possible from the corrupted message.

ENCODERS AND DECODERS

Encoder:

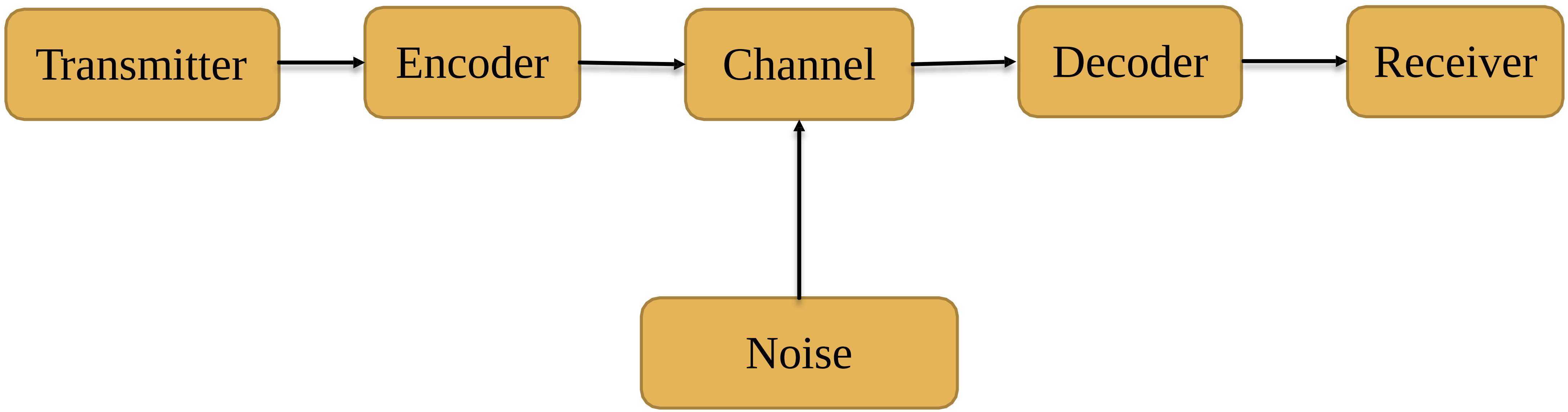
An encoder is a device which transforms the incoming messages in such a way that the presence of noise in the transformed messages is detectable.

Decoder:

A decoder is a device which transforms the encoded message into their original form that can be understood by the receiver.

- So, by using a suitable encoder and decoder, it may he possible to detect the distortions in the messages due to noise in the channel and to correct them.

TYPICAL DATA COMMUNICATION MODEL WITH NOISE



EXPLANATION

- The input message can consist of a sequence of letters, characters or symbol set called alphabet.
- The input message will be transformed by the encoder into a string of characters or symbols of another alphabet in a one-to-one fashion.
- We will discuss a binary channel in which the encoder will transform an input message into a binary string consisting of the symbols 0 and 1.
- Decoding can be seen as the inverse operation of encoding.

GROUP CODE

Definition:

If $B = \{0,1\}$ then $B^n = \{x_1, x_2, x_3, \ldots x_n \mid x_i \in B, i=1,2,3, \ldots n\}$ is a group under the binary operation of addition modulo 2 (denoted by $\oplus$), then $(B^n, \oplus)$ is called a **group code**.

HAMMING CODES

Definition:

- The codes obtained by introducing additional digits called *parity digits* to the digits in the original message are called **hamming codes**.

In Hamming's single-error detecting code, the last digit is made either 0 or 1 and the remaining digits contain the information part of the message.

If the digit introduced in the last position gives an even number/odd number of 1's in the encoded word, the extra digit is called an even/ odd **parity check**.

EXAMPLE OF PARITY CHECK

Code before parity check	Code after even parity check	Code after odd parity check
000	0000	0001
001	0011	0010
010	0101	0100
011	0110	0111
100	1001	1000
101	1010	1011
110	1100	1101
111	1111	1110

HAMMING DISTANCE

Definition:

- Suppose if x and y represents binary strings $x_1x_2x_3 \dots x_n$ and $y_1y_2y_3 \dots y_n$ then the number of positions in the strings for which $x_i \neq y_i$ is called **Hamming Distance** between x and y.
- Denoted by H(x,y)

$$H(x,y) = \text{weight of } x \oplus y = |\, x \oplus y \,|$$

BASIC NOTATIONS OF ERRORS CORRECTION USING MATRICES

- Where $m,n \in Z^+$, $(m < n)$ {m should be strictly less than n}, the coding function $e: B^m \rightarrow B^n$ [Where B = (0,1)] is given by a mxn matrix G over B.
- The matrix G is called as generator matrix for the code and is of the form $[I_m|A]$ where I_m is the mxn unit matrix and A is an m*(n-m) matrix chosen suitably.

$$H= [A^T|\ I_{n-m}]$$

- If w is a message $\in B^m$ then e(w) = wG and the code (the set of code words) $c = e(B^m) \subseteq B^n$, where w is a (1xm) vector.

EXAMPLE

- Let w ∈ B^2 and G is $\begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$
- The words belonging to B^2 are 00, 01, 10, 11. Then we have the code words as:

$$e(00) = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$e(01) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \end{bmatrix}$$

$$e(10) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

$$e(11) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

- Question. Find the code words generated by the encoding function $e: B^m \rightarrow B^n$ with respect to the parity check matrix given by

$$H = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution:

Let $e = B^m \rightarrow B^n$ where $m < n$ be the encoding function.
Here, m=2 and n =5

G = [I$_m$ | A] be the generated matrix, So, we have $G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$

Now, $B^2 = \{00, 01, 10, 11\}$

From **e(w) = wG** we have :

$$e(00) = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$e(01) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$e(10) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$$e(11) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

So, the code words generated are 00000, 01011, 10011, 11000

- Question. Find the code words generated by the parity check matrix given by

$$H = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

when the encoding function is $e: B^3 \rightarrow B^6$.

Solution:

Let $e = B^m \rightarrow B^n$ where $m < n$ be the encoding function.
Here, $m=3$ and $n=6$

$G = [I_m \,|\, A]$ be the generated matrix, So, we have $G = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$

Now, $B^3 = \{000, 001, 010, 100, 101, 110, 111\}$

From $e(w) = wG$ we have :

$$e(000) = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

solution

$$e(001) = \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$$e(010) = \begin{pmatrix} 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

$$e(011) = \begin{pmatrix} 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 \end{pmatrix}$$

$$e(100) = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$e(101) = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$e(110) = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$e(111) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \begin{pmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Therefore, the code words generated are
000000, 001011, 010101, 011110, 100111, 101100, 110010,111001

PARITY CHECK MATRIX

- Parity check matrix is used for decoding at receiver.
- Parity check matrix is denoted by H
- $H = [A^T \mid I_{n-k}]$
- Where A is same parity bits matrix that is used to construct
  G(generator matrix) for encode and I is identity matrix.

ERROR DETECTION USING PARITY CHECK MATRIX

Given, r: which is received word

H : unique parity check matrix which corrects the single error in transmission.

Three cases:

1. $H.\ R^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

There is no error in transition and r is the code word transmitted.

ERROR DETECTION USING PARITY CHECK MATRIX

2. If H. R^t = ith column of H, then we conclude that there is a single error in r during transmission and it has occurred in the ith component of r. Changing the ith component of r, we get the code word c.

$$H.R^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

ERROR DETECTION USING PARITY CHECK MATRIX

3. If **neither case (i) nor case (ii) occurs we conclude that more than one transmission error have occurred. Detection is possible in this case but correction is not possible.**

H. R^t is not equal to any column of H

$$H.\,R^t = \begin{bmatrix} 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

PRACTICE QUESTIONS

Q1. Given the generator matrix, G= $\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$

Corresponding to the encoding function, e: $B^3 \rightarrow B^6$, find the corresponding parity check matrix and use it to decode the following received words and hence, to find the original message. Are all words decoded uniquely?

(i) 110101 (ii) 001111 (iii) 111111

PRACTICE QUESTIONS

solution : we assume

$G= [I_3 \mid A]$

Then, H=

$[A^T \mid I_3]$ H = $\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix}$

(i) $H. r^T$ = $\begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Since it is 0, the received word in this case is the transmitted word itself. Original message is 110.

PRACTICE QUESTIONS

(ii)

$$H.\, r^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

Since $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ is the fifth column of H, the element in the fifth position of r is changed.

The decoded word is 001101 and original message is 001.

PRACTICE QUESTIONS

(iii)

$$H.\, r^T = \begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Since, the syndrome is not identical with any column of H, the received word cannot be decoded uniquely.

PRACTICE QUESTIONS

Ques 2. Decode each of the following received words corresponding to the encoding function e: $B^3 \to B^6$ given by
e(000)=000000,e(001)=001011,e(010)=010101,e(100)=100111,e(011)=011110,e(101)=101100,e(110)=110010 and e(111)=111000, assuming that no error or signal error has occurred:

011110,110111,110000

PRACTICE QUESTIONS

(i) The word 011110 is identical with e(011).hence no error has occurred and original message is 011.

(ii) 110111 differs from e(100)=100111 in second position only. Transmitted message is 100111 and original message is 100.

(iii) 110000 differs from e(110)=110010 in fifth position only. Transmitted message is 110010 and original message is 110.